1

$$A = \left(\begin{array}{ccc|ccc} 3 & 1 & 1 & 1 & 0 & 0 \\ 1 & 3 & 1 & 0 & 1 & 0 \\ 1 & 1 & 3 & 0 & 0 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|ccc} 3 & 1 & 1 & 1 & 0 & 0 \\ 0 & 8 & 2 & -1 & 3 & 0 \\ 0 & 2 & -2 & 0 & 1 & -1 \end{array}\right)$$

$$\left(\begin{array}{ccc|ccc} 3 & 1 & 1 & 1 & 0 & 0 \\ 0 & 8 & 2 & -1 & 3 & 0 \\ 0 & 0 & 10 & -1 & -1 & 4 \end{array}\right)$$

$$\left(\begin{array}{ccc|ccc} 30 & 10 & 0 & 11 & 1 & -4 \\ 0 & 40 & 0 & -4 & 16 & -4 \\ 0 & 0 & 10 & -1 & -1 & 4 \end{array}\right)$$

$$\left(\begin{array}{ccc|ccc} 120 & 0 & 0 & 45 & -12 & -12 \\ 0 & 40 & 0 & -4 & +16 & -4 \\ 0 & 0 & 10 & -1 & -1 & 4 \end{array}\right)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & .4 & -.1 & -.1 \\ 0 & 1 & 0 & -.1 & .4 & -.1 \\ 0 & 0 & 1 & -.1 & -.1 & .4 \end{array}\right)$$

$$\mathrm{Det}(A) = 3(9-1) - 1(3-1) + 1(1-3)$$

$$= \underline{\underline{20}}$$

$$\left(\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 2 & 3 & 4 & 0 & 1 & 0 & 0 \\ 1 & 3 & 6 & 10 & 0 & 0 & 1 & 0 \\ 1 & 4 & 10 & 20 & 0 & 0 & 0 & 1 \end{array}\right)$$

$$\left(\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 & -1 & 1 & 0 & 0 \\ 0 & 2 & 5 & 9 & -1 & 0 & 1 & 0 \\ 0 & 3 & 9 & 19 & -1 & 0 & 0 & 1 \end{array}\right)$$

$$\left(\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 & 1 & -2 & 1 & 0 \\ 0 & 0 & 3 & 10 & 2 & -3 & 0 & 1 \end{array}\right)$$

$$\left(\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 3 & -3 & 1 \end{array}\right)$$

$$\left(\begin{array}{cccc|cccc} 1 & 0 & -1 & -2 & 2 & -1 & 0 & 0 \\ 0 & 1 & 0 & -3 & -3 & 5 & -2 & 0 \\ 0 & 0 & 1 & 0 & 4 & -11 & 10 & -3 \\ 0 & 0 & 0 & 1 & -1 & 3 & -3 & 1 \end{array}\right)$$

$$\left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 4 & -6 & 4 & -1 \\ 0 & 1 & 0 & 0 & -6 & -14 & -11 & 3 \\ 0 & 0 & 1 & 0 & 4 & -4 & 10 & -3 \\ 0 & 0 & 0 & 1 & -1 & 3 & -3 & 1 \end{array}\right)$$

$\mathrm{Det} = \underline{1}$

2. Let A & B are square upper triangular matrix

I- SUM

Let sum of A and B = C

So, $\cancel{C_{ij}} = \cancel{A_{ij}} + \cancel{B_{ij}}$

$c_{ij} = a_{ij} + b_{ij}$ [elements of respective matrix]

a_{ij} and $b_{ij} = 0$ $[i > j]$

$\Rightarrow c_{ij} = 0$ $[i > j]$

proved

II. PRODUCT

Let product of A and B = D

$d_{ij} = \Sigma a_{ik} \cdot b_{kj}$

d_{ij} should be zero for $i > j$ — (1)

Now, a_{ik} and $b_{kj} \neq 0$ $[i <= k, k <= j]$

$\Rightarrow [i <= j]$ — (2)

(1) & (2) contradicts

$\Rightarrow$ at least, either $a_{ij} = 0$ or $b_{ij} = 0$

proved

III: INVERSE

Since, $A^{-1} = \frac{1}{\det(A)} \cdot adj(A)$

Matrix of cofactor = C

$C_{ij} = (-1)^{i+j} M_{ij}$, where $M_{ij} = 0$ $[i < j]$

$\Rightarrow C_{ij} = 0$ $(i < j)$ and $adj(A)_{ij} = 0$ $[i > j]$

$\Rightarrow adj(A)$ is upper triangular matrix. proved

3 a) $L_1 U_1 = L(D)(D^{-1} U)$

$= L(D \cdot D^{-1}) U$

$= L U$

$= A$

$L_1 U_1$ is also a factorisation of A

b) $L_1 U_1 = L_2 U_2$

$\Rightarrow L_1 U_1 U_1^{-1} = L_2 U_2 U_1^{-1}$

$L_1 = L_2 D$

also,

$L_1^{-1} L_1 U_1 = L_1^{-1} L_2 U_2$

$U_1 = D^{-1} U_2$